

ELEC30 - Laboratório de Controle Discreto

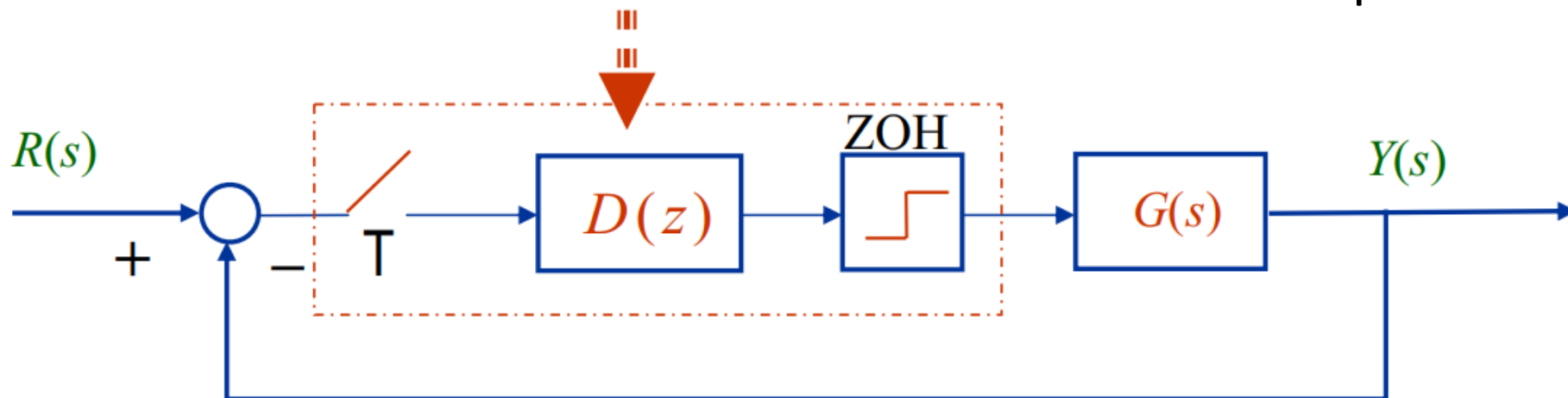
Aula 5: Mapeamento entre os planos “s” e “z”

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Sistema de Controle Discreto: Primeiros passos



- Escolher o período de amostragem T , de acordo com as especificações do sistema contínuo $G(s)$.
- Calcular o equivalente discreto $G(z)$, incluindo as partes contínua/discreta do grameador, a partir da transformada de Laplace inversa:

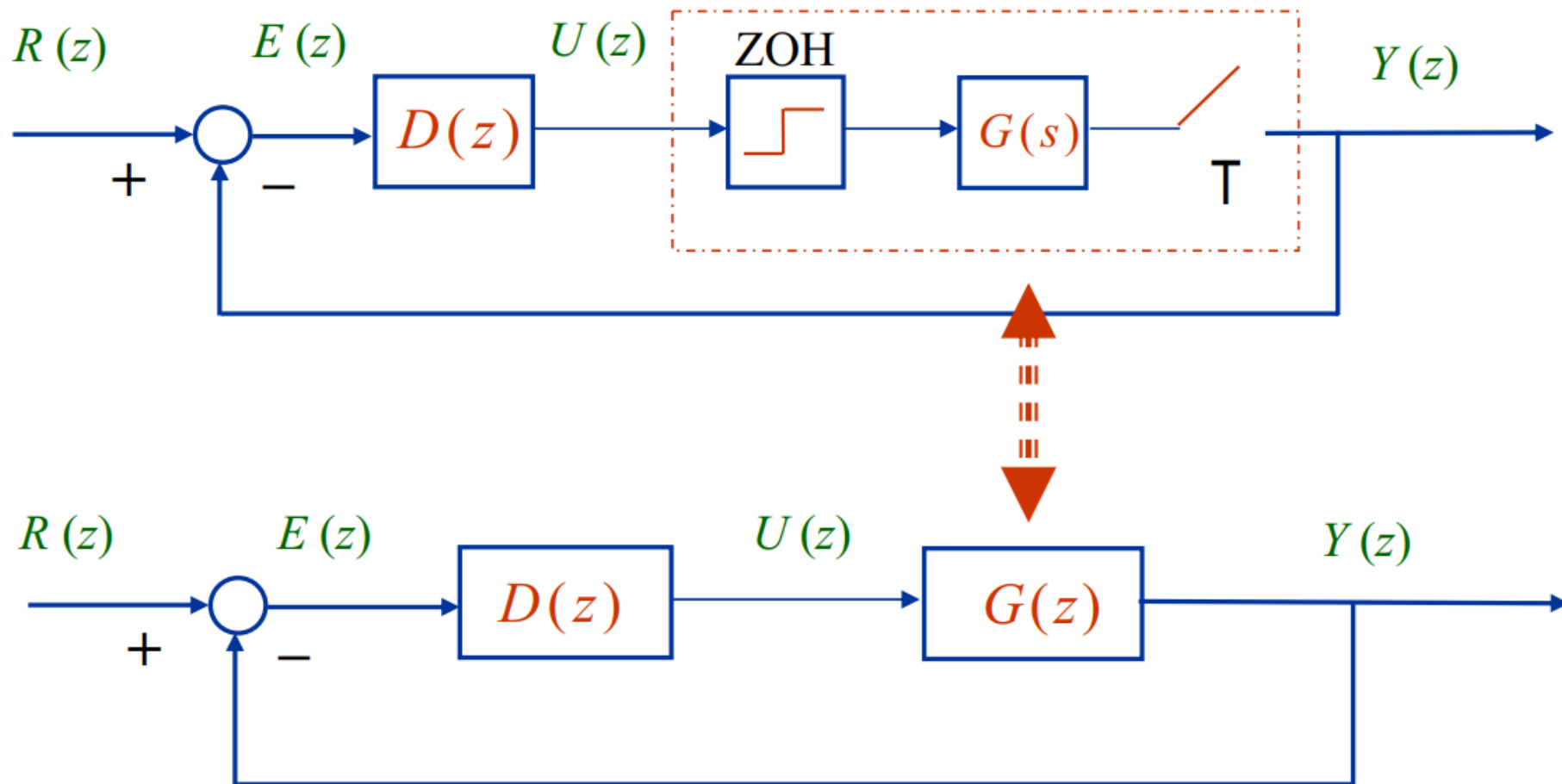
$$G(z) = Z \left\{ \mathcal{L}^{-1} \left\{ \frac{G(s)}{s} \right\} \right\} \cdot \frac{z-1}{z}$$

- A inclusão de $G(z)$ na malha de controle permite projetar e analisar o sistema de controle completo utilizando a transformada Z .

Tabela de Transformada Z: sinais mais frequentes

	$X(s)$	$x(t)$	$X(kT)$	$X(z)$
1	1	$\delta(t)$	$\delta(kT)$	1
2	$\frac{1}{s}$	1(t)	1(kT)	$\frac{1}{1 - z^{-1}}$
3	$\frac{1}{s + a}$	e^{-at}	e^{-akT}	$\frac{1}{1 - e^{-aT} z^{-1}}$
4	$\frac{1}{s^2}$	t	kT	$\frac{Tz^{-1}}{(1 - z^{-1})^2}$
5	$\frac{\omega}{s^2 + \omega^2}$	$\sin(\omega t)$	$\sin(\omega kT)$	$\frac{z^{-1} \sin \omega T}{1 - 2z^{-1} \cos \omega T + z^{-2}}$
6	$\frac{s}{s^2 + \omega^2}$	$\cos(\omega t)$	$\cos(\omega kT)$	$\frac{1 - z^{-1} \cos \omega T}{1 - 2z^{-1} \cos \omega T + z^{-2}}$

Sistema de Controle Discreto: Primeiros passos



Discretização de uma FT

- Dado uma planta contínua, $G(s)$.

$$y(t) = G(s)u(t) = \frac{B(s)}{A(s)}u(t) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \dots + a_0}u(t)$$

- Obtém-se a equação diferencial ligando suas entradas e saídas:

$$\frac{d^n}{dt^n}y(t) + a_{n-1}\frac{d^{n-1}}{dt^{n-1}}y(t) + \dots + a_0y(t) = b_{n-1}\frac{d^n}{dt^n}u(t) + \dots + b_0yu(t)$$

Como obter a função de transferência discreta $G(z)$ a partir de uma função de transferência contínua $G(s)$?

Discretização de uma FT

- A solução analítica é obtida fazendo $z = e^{sT}$ na função de transferência $G(s)$:

$$G(z) = G(s) \big|_{e^{sT}=z}$$

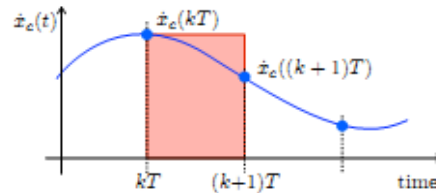
- Do ponto de vista computacional, a discretização no tempo do sinal $\mathbf{y(t)=G(s)u(t)}$ nada mais é do que uma **aproximação numérica** (com **passo de integração** constante **T**), que gera as amostras de saídas a partir das amostras de entrada :

$$y(kT) \triangleq \tilde{y}(k) = \tilde{G}(z)\tilde{u}(k) \triangleq G(z)u(kT)$$

- Existem vários métodos numéricos para resolver equações diferenciais (passo constante, passo variável, aproximação de ordem n, etc)

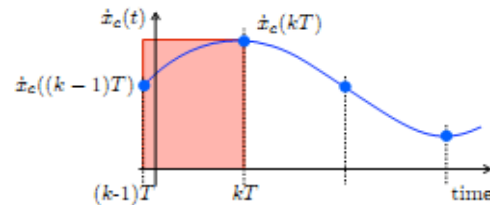
Simulação de sistemas contínuos: Integração

- *(forward) Euler method*



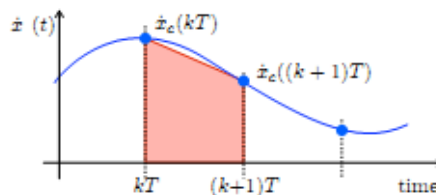
$$\underbrace{x_c((k+1)T) - x_c(kT)}_{(z-1)x_c(kT)} = \int_{kT}^{(k+1)T} \dot{x}_c(\tau) d\tau \approx \underbrace{T \dot{x}_c(kT)}_{Tsx_c(kT)} \rightarrow s = \frac{z-1}{T}$$

- *backward Euler method*



$$\underbrace{x_c(kT) - x_c((k-1)T)}_{(1-z^{-1})x_c(kT)} = \int_{(k-1)T}^{kT} \dot{x}_c(\tau) d\tau \approx \underbrace{T \dot{x}_c(kT)}_{Tsx_c(kT)} \rightarrow s = \frac{1-z^{-1}}{T}$$

- *Trapezoidal rule*



$$\underbrace{x_c((k+1)T) - x_c(kT)}_{(z-1)x_c(kT)} = \int_{kT}^{(k+1)T} \dot{x}_c(\tau) d\tau \approx \underbrace{\frac{T[\dot{x}_c((k+1)T) + \dot{x}_c(kT)]}{2}}_{\frac{T}{2}(z+1)sx_c(kT)}$$

$$\rightarrow s = \frac{2(z-1)}{T(z+1)}$$

Relação entre o plano “s” e o plano “z”

- Existe uma relação entre a localização de um polo de $G(s)$ e o seu equivalente discreto no plano z , isto é, os polos de $G(z)$.

- forward Euler differences:

$$\frac{z_i - 1}{T} = s_i \Rightarrow z_i = 1 + Ts_i$$

- backward Euler differences:

$$\frac{z_i - 1}{Tz_i} = s_i \Rightarrow z_i = \frac{1}{1 - s_i T}$$

- Tustin's method:

$$\frac{2(z_i - 1)}{T(z_i + 1)} = s_i \Rightarrow z_i = \frac{1 + s_i T/2}{1 - s_i T/2}$$

- Exact method: $z_i = e^{s_i T}$

Plano “s”



Plano “z”

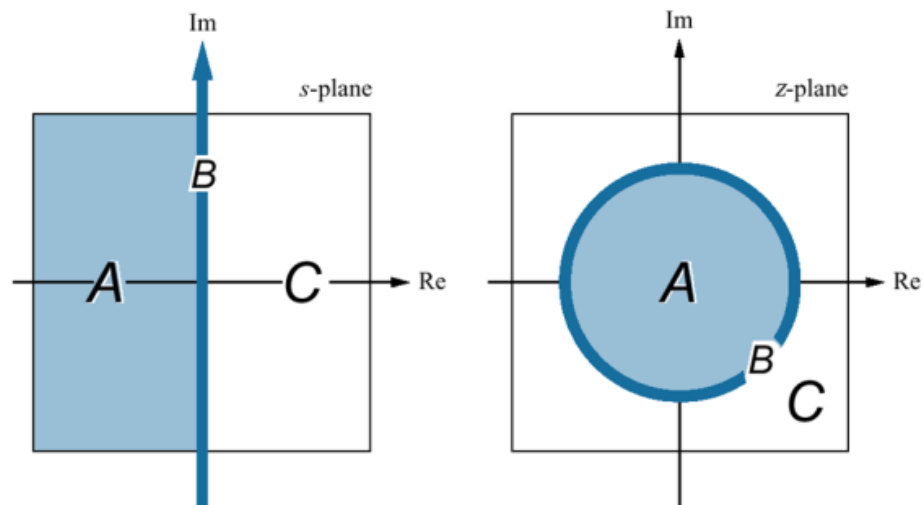
$$\begin{aligned} z &= e^{sT} \\ &= e^{T(\alpha + j\omega)} \\ &= e^{\alpha T} e^{j\omega T} \\ &= e^{\alpha T} \angle \omega T \end{aligned}$$

$$z = e^{\alpha T} \angle \omega T$$

$$\alpha > 0 \rightarrow e^{\alpha T} > 1$$

$$\alpha = 0 \rightarrow e^{\alpha T} = 1$$

$$\alpha < 0 \rightarrow e^{\alpha T} < 1$$



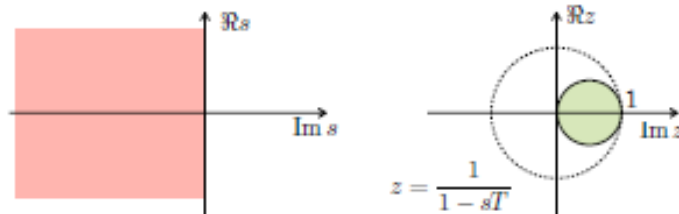
Relação entre o plano “s” e o plano “z”

forward Euler
differences



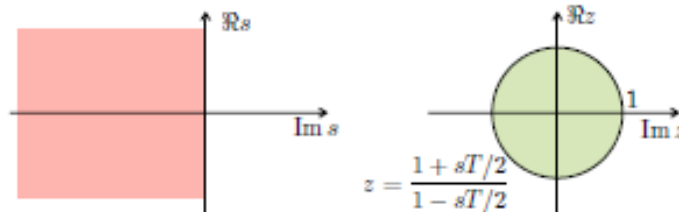
stable poles in s may be
mapped to unstable poles
in z

backward Euler
differences



stable poles in s are also
stable poles in z ,
marginally stable poles
may become
asymptotically stable

Tustin's rule



stable poles in s are also
stable poles in z

Exemplo de discretização de $G(s)$

- Consider the continuous-time controller

$$u(t) = \frac{1}{s+2}e(t) \Rightarrow \frac{d}{dt}u(t) + 2u(t) = e(t)$$

- Use *Euler method* to approximate the derivative

$$\frac{d}{dt}u(t) \approx \frac{u((k+1)T) - u(kT)}{T} = \frac{1}{T}(\tilde{u}(k+1) - \tilde{u}(k))$$

- Recall that z represents the unit-shift operator $z\tilde{u}(k) = \tilde{u}(k+1)$

$$\tilde{u}(k) = \frac{1}{\left(\frac{z-1}{T}\right) + 2}\tilde{e}(k) \triangleq \tilde{C}(z)\tilde{e}(k)$$

- Note that formally this is equivalent to replace $s = \left(\frac{z-1}{T}\right)$ in $C(s)$ to get $\tilde{C}(z)$

Plano “s”



Plano “z”

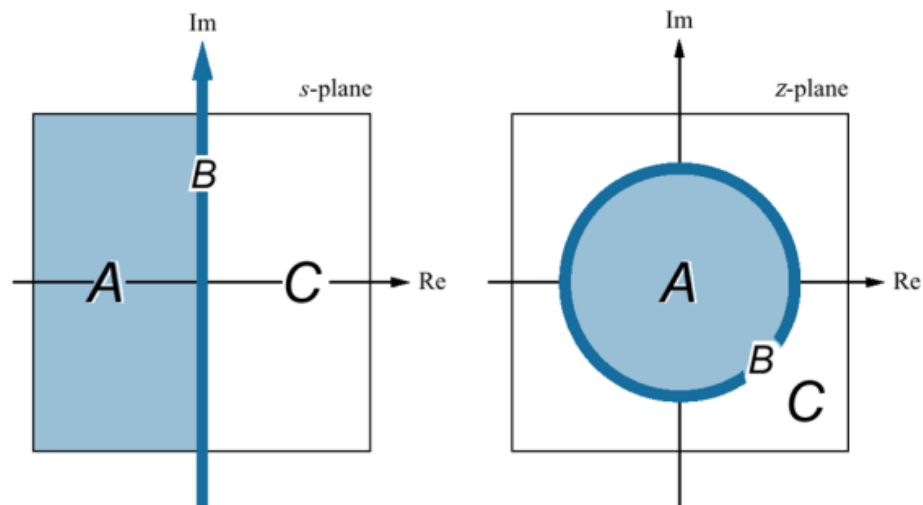
$$\begin{aligned} z &= e^{sT} \\ &= e^{T(\alpha + j\omega)} \\ &= e^{\alpha T} e^{j\omega T} \\ &= e^{\alpha T} \angle \omega T \end{aligned}$$

$$z = e^{\alpha T} \angle \omega T$$

$$\alpha > 0 \rightarrow e^{\alpha T} > 1$$

$$\alpha = 0 \rightarrow e^{\alpha T} = 1$$

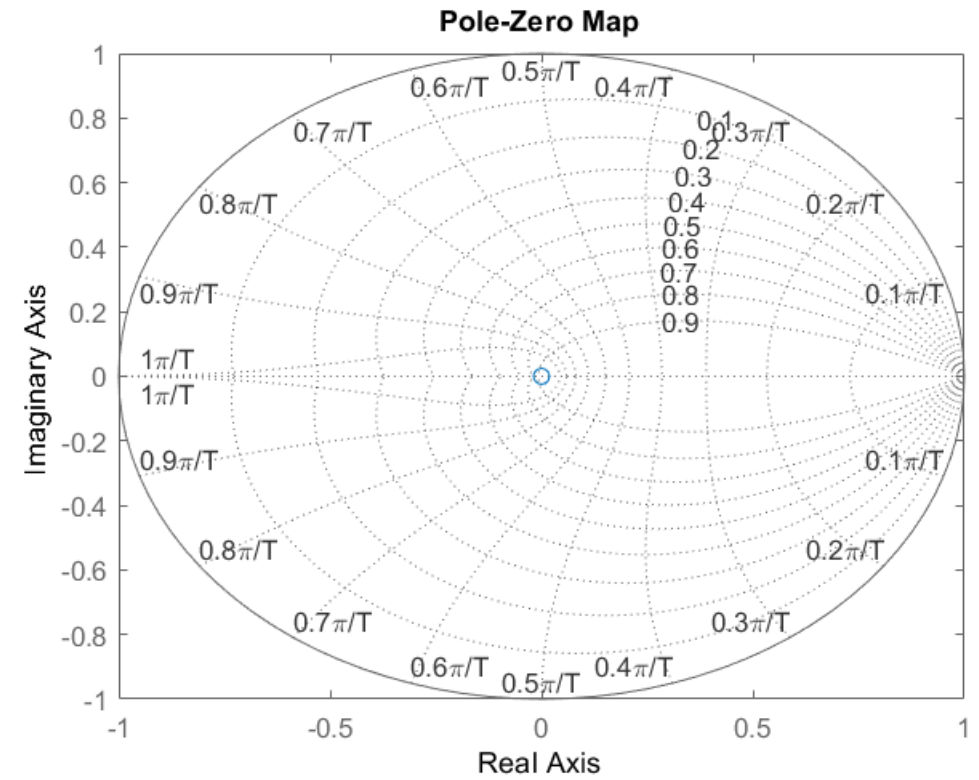
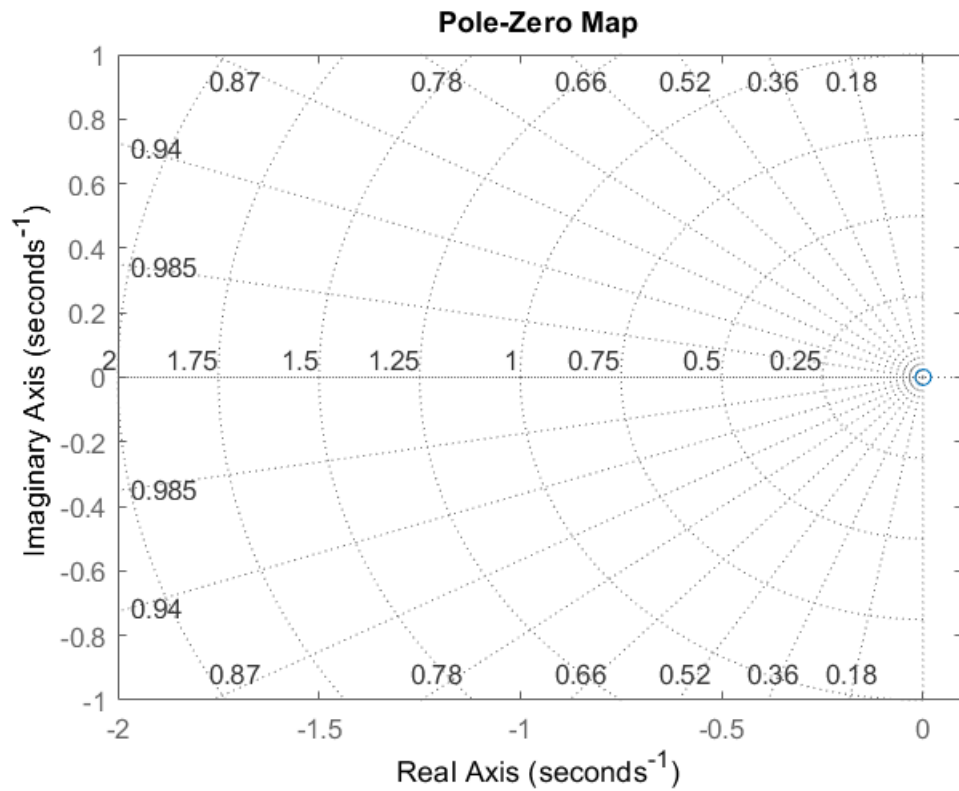
$$\alpha < 0 \rightarrow e^{\alpha T} < 1$$



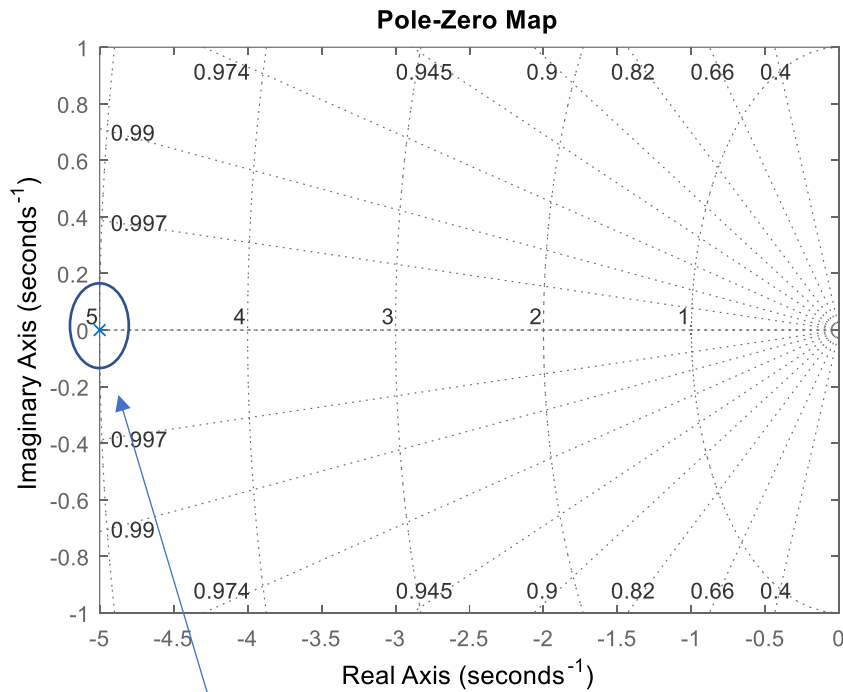
Plano “s”



Plano “z”



Efeito da amostragem



$$G(s) = \frac{5}{s + 5}$$

$$T = 1s$$

$$T = 0.1s$$

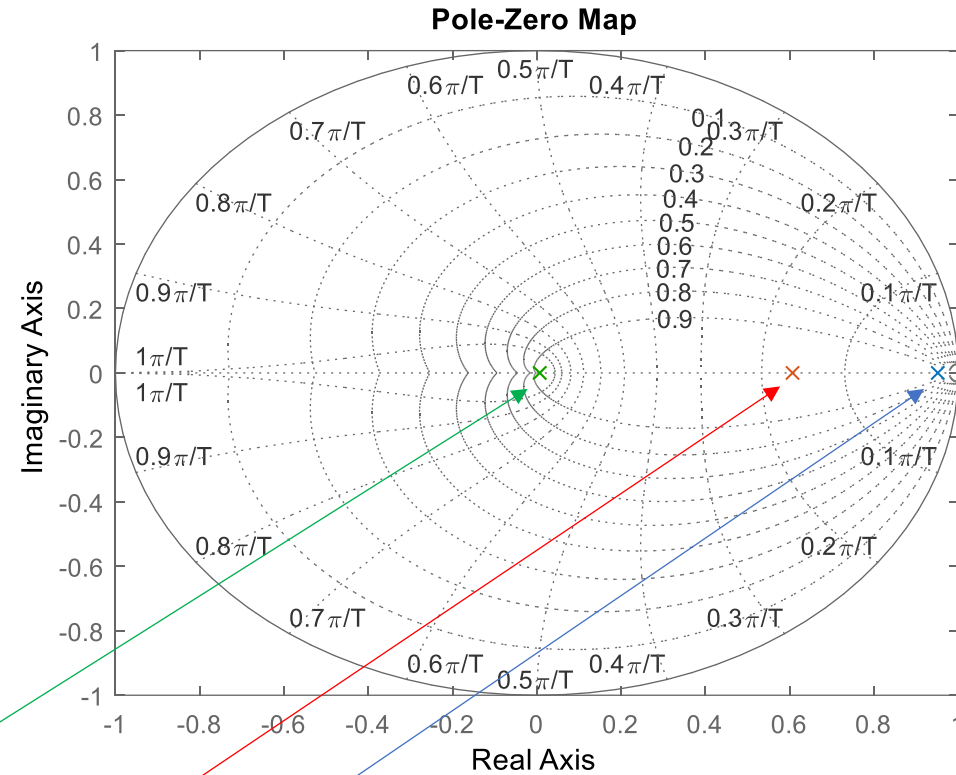
$$T = 0.01s$$

Com ZOH

$$G1(z) = \frac{0.9933}{z - 0.006738}$$

$$G2(z) = \frac{0.3935}{z - 0.6065}$$

$$G3(z) = \frac{0.04877}{z - 0.9512}$$



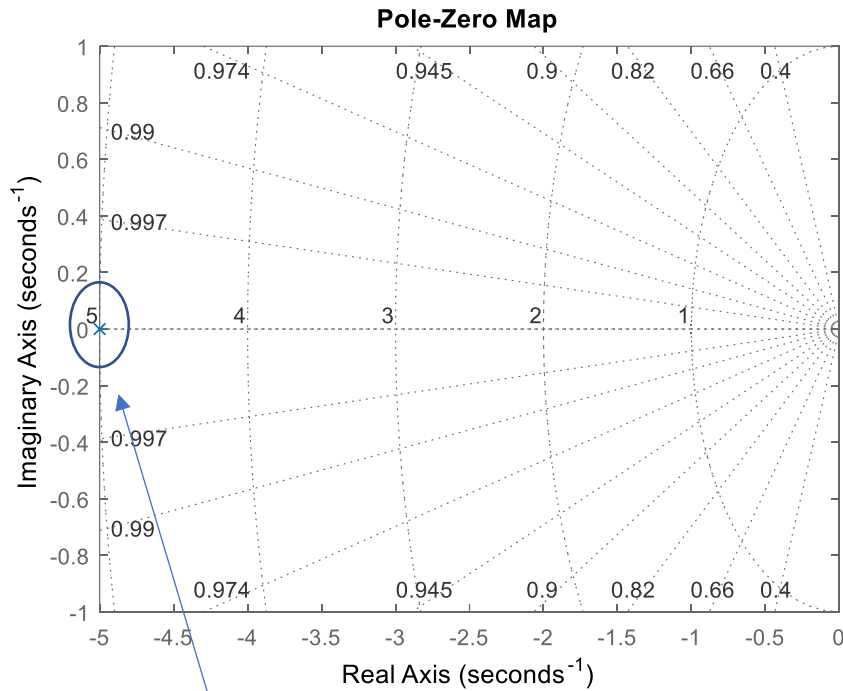
$$p_s = -5 \Rightarrow p_z = e^{p_s T}$$

$$p_z = e^{-5} = 0.0067$$

$$p_z = e^{-0.5} = 0.6065$$

$$p_z = e^{-0.05} = 0.9512$$

Efeito do segurador



$$G(s) = \frac{5}{s + 5}$$

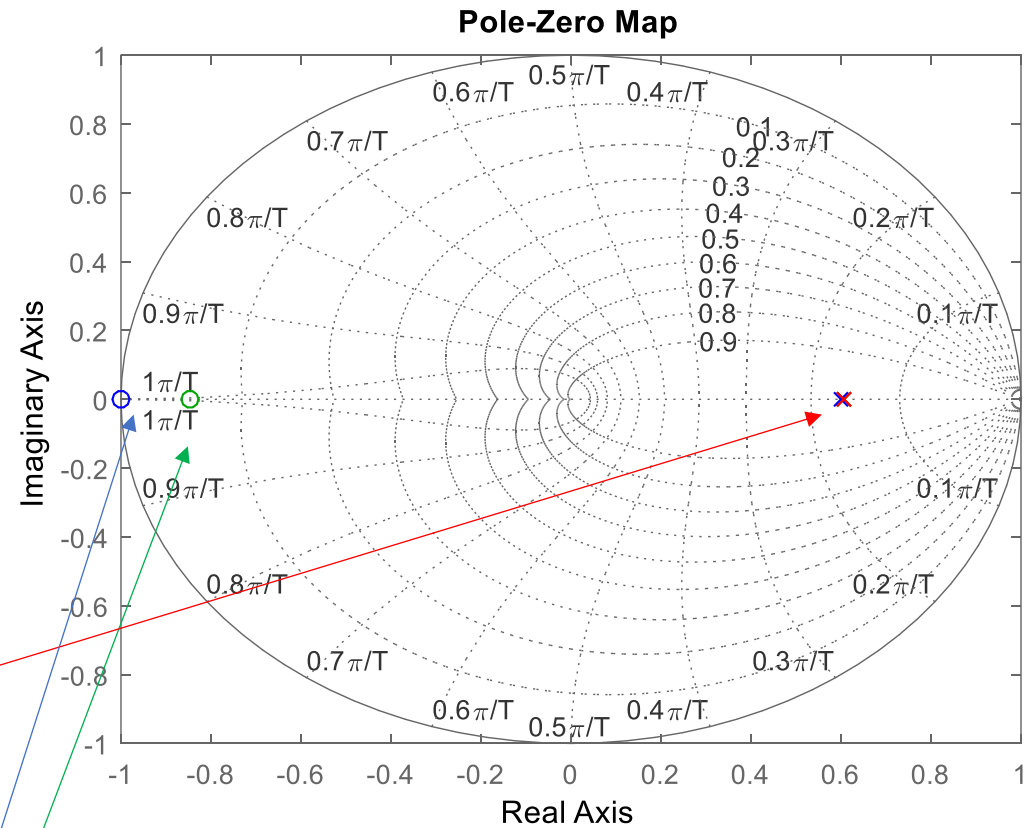
ZOH
FOH
TUSTIN

$T=0.1s$

$$G2(z) = \frac{0.3935}{z - 0.6065}$$

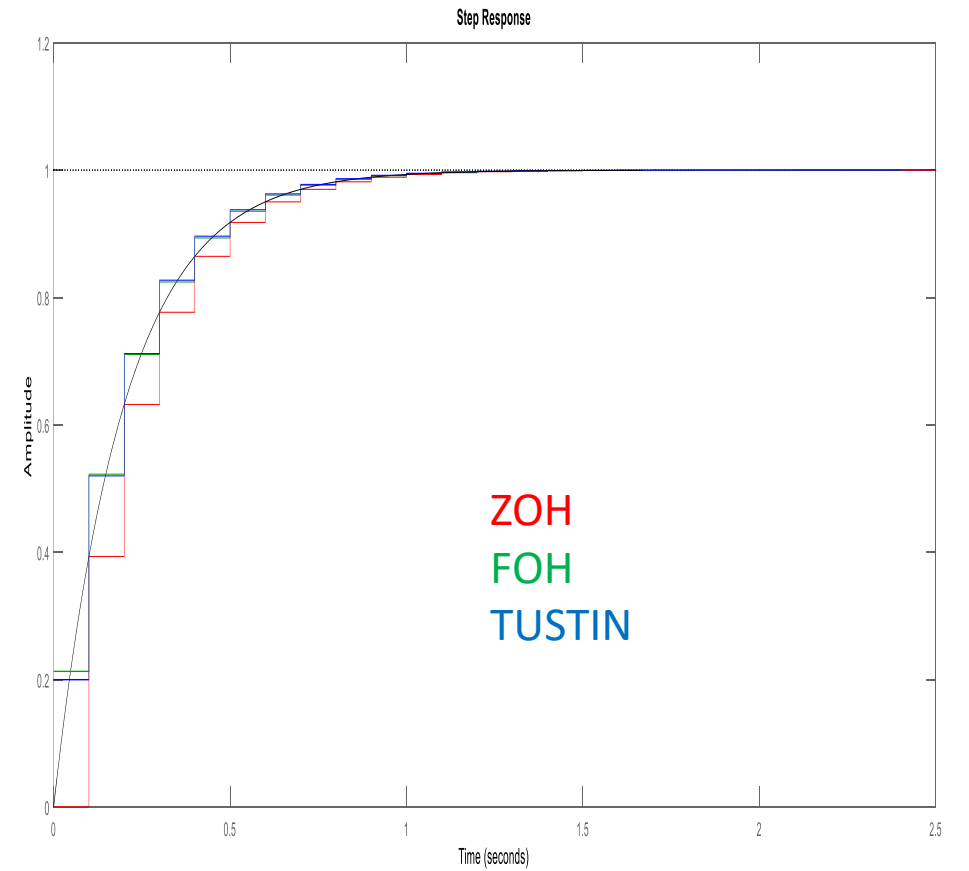
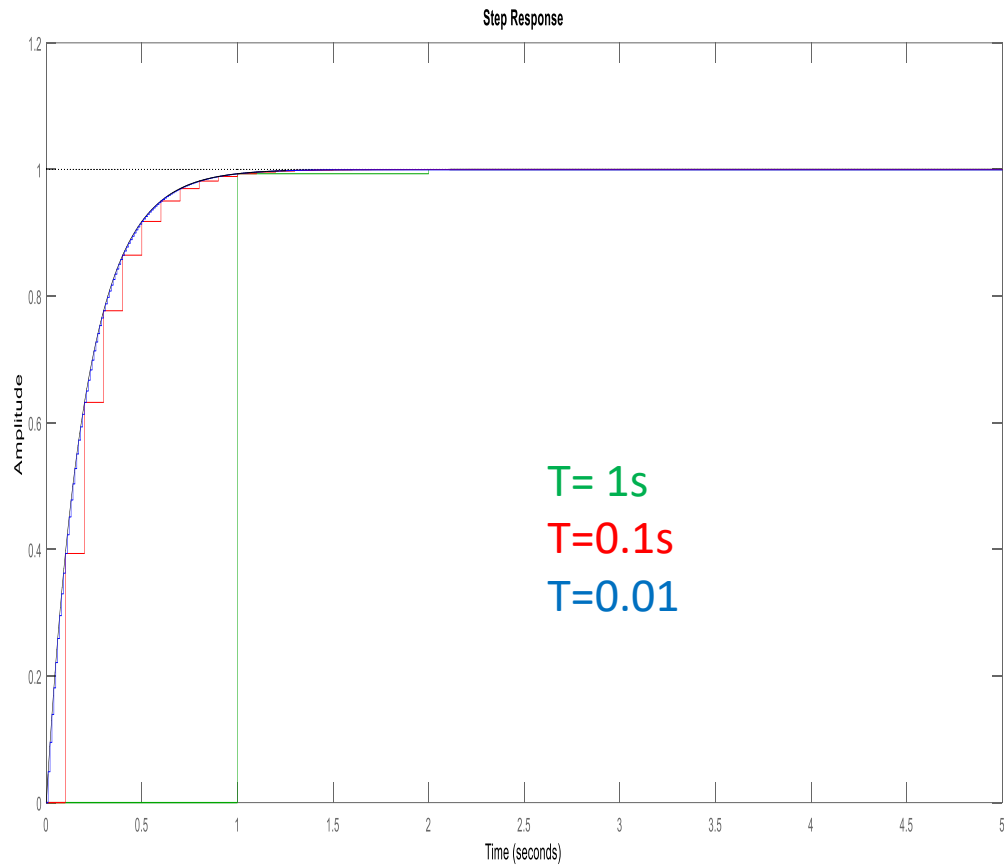
$$G2(z) = \frac{0.2131z + 0.1804}{z - 0.6065}$$

$$G3(z) = \frac{0.2z + 0.2}{z - 0.6}$$

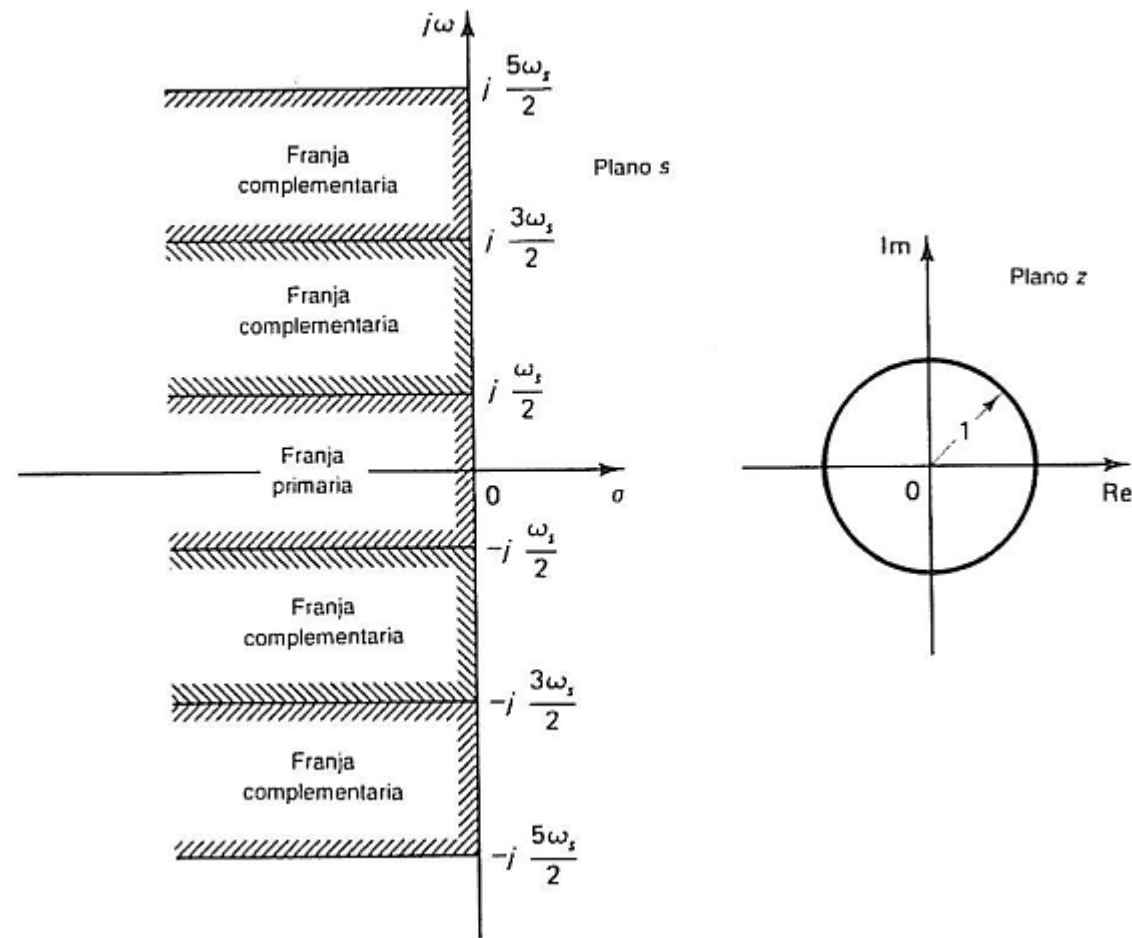


- Todos tem o mesmo polo
- FOH e TUSTIN acrescentam zeros

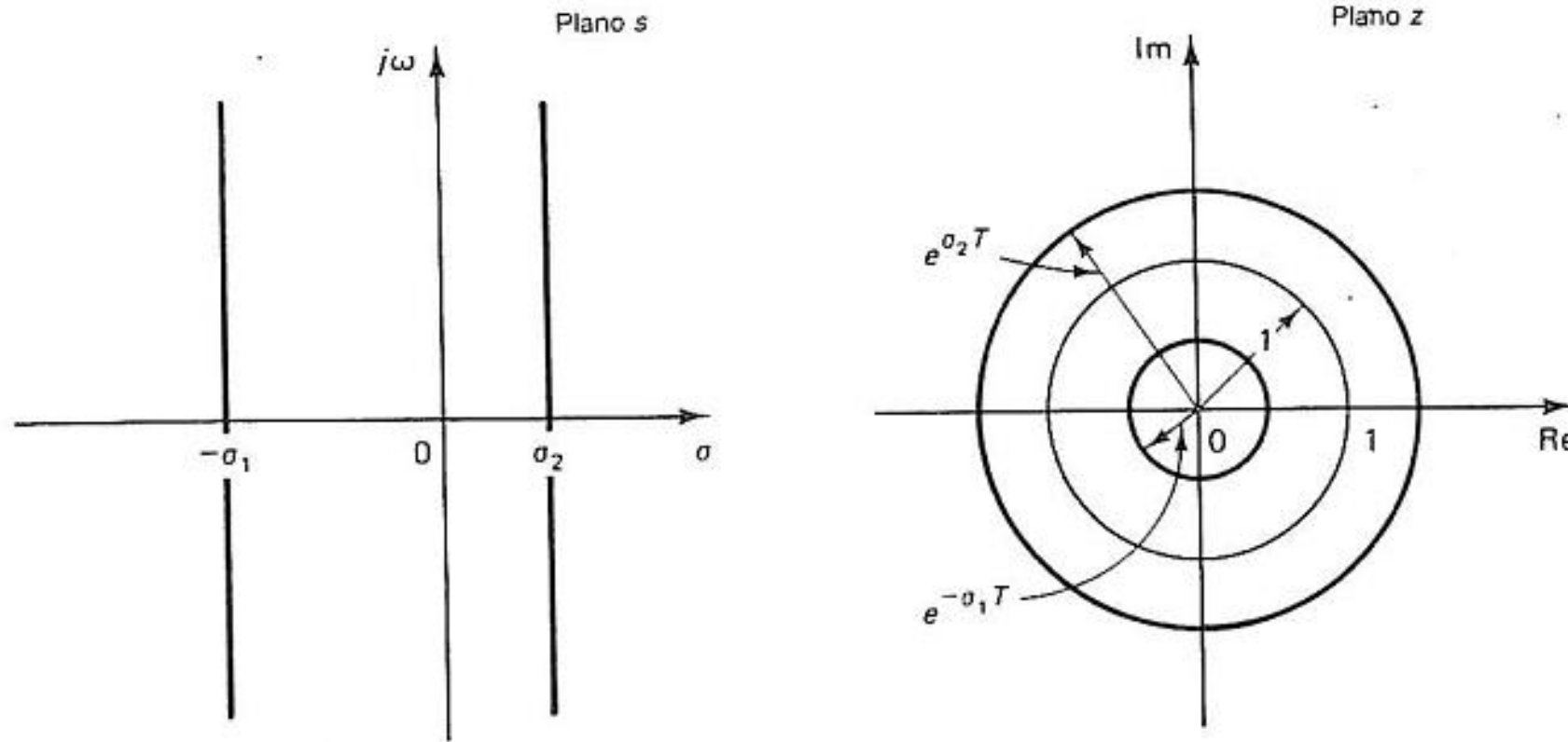
Resposta Temporal



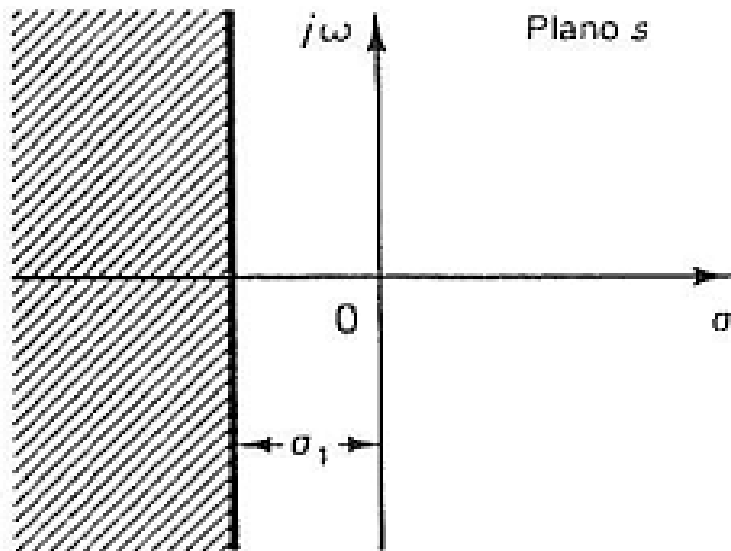
Correspondência entre o plano s e o plano z



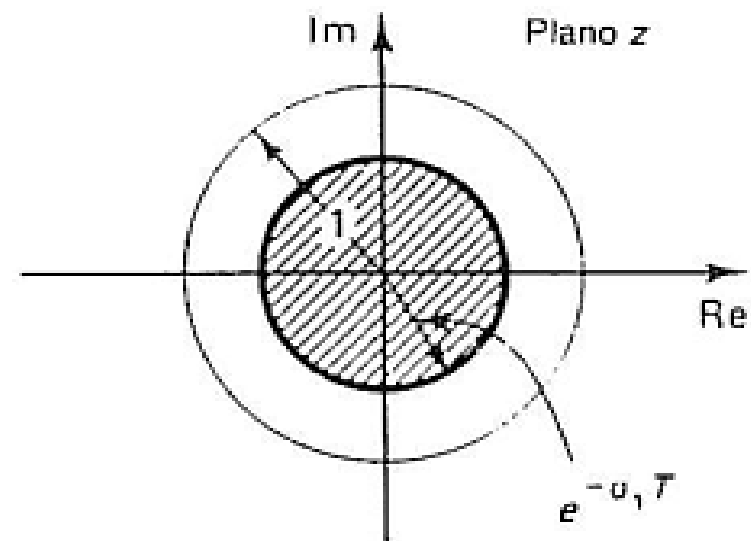
Constante de tempo



Tempo de acomodação $< \sigma_1$

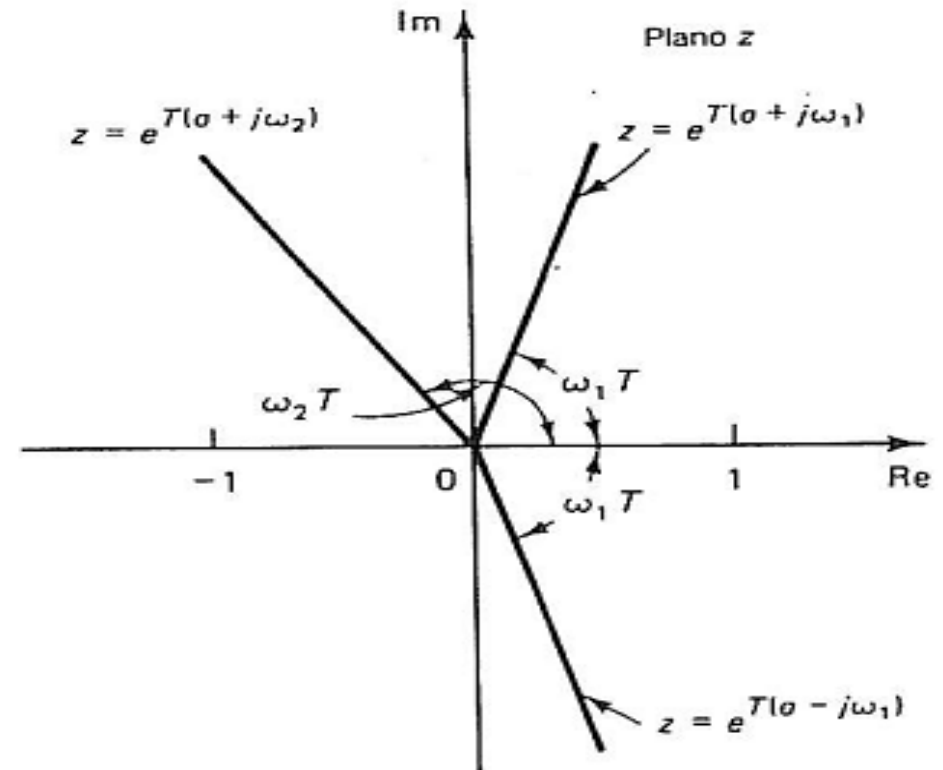
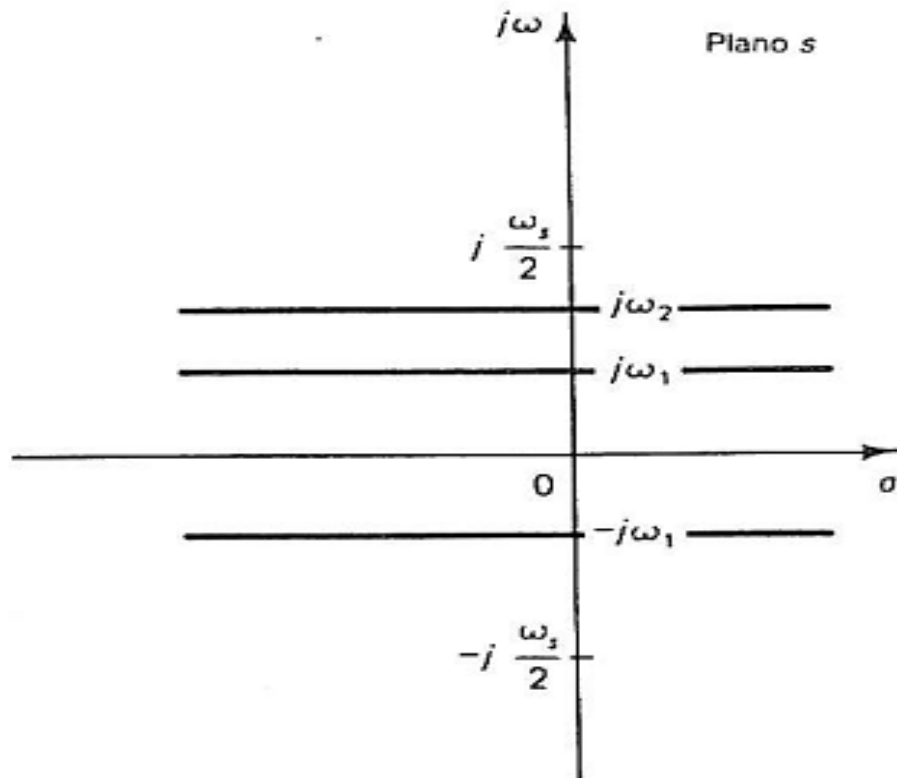


a)

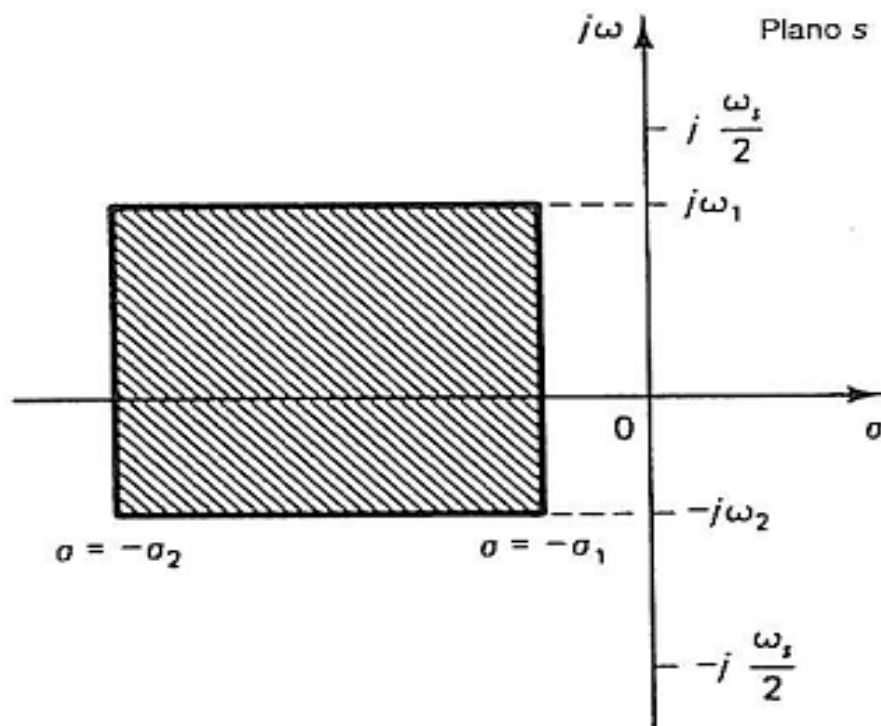


b)

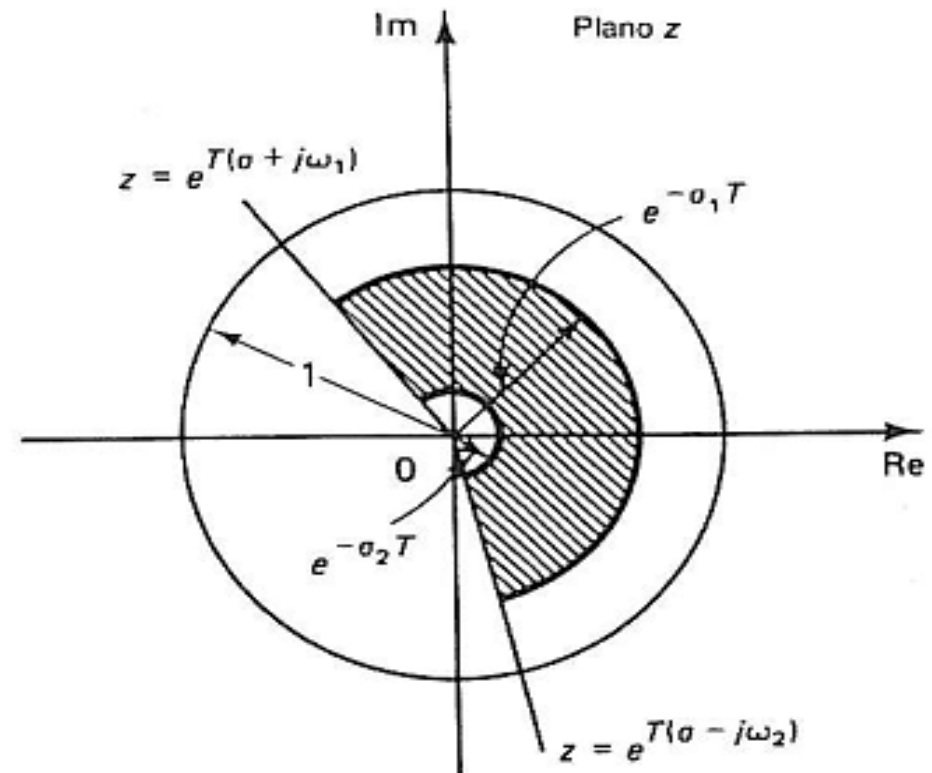
Frequencia constante



Lugar geométrico s e z



a)



b)

Fator de amortecimento constante

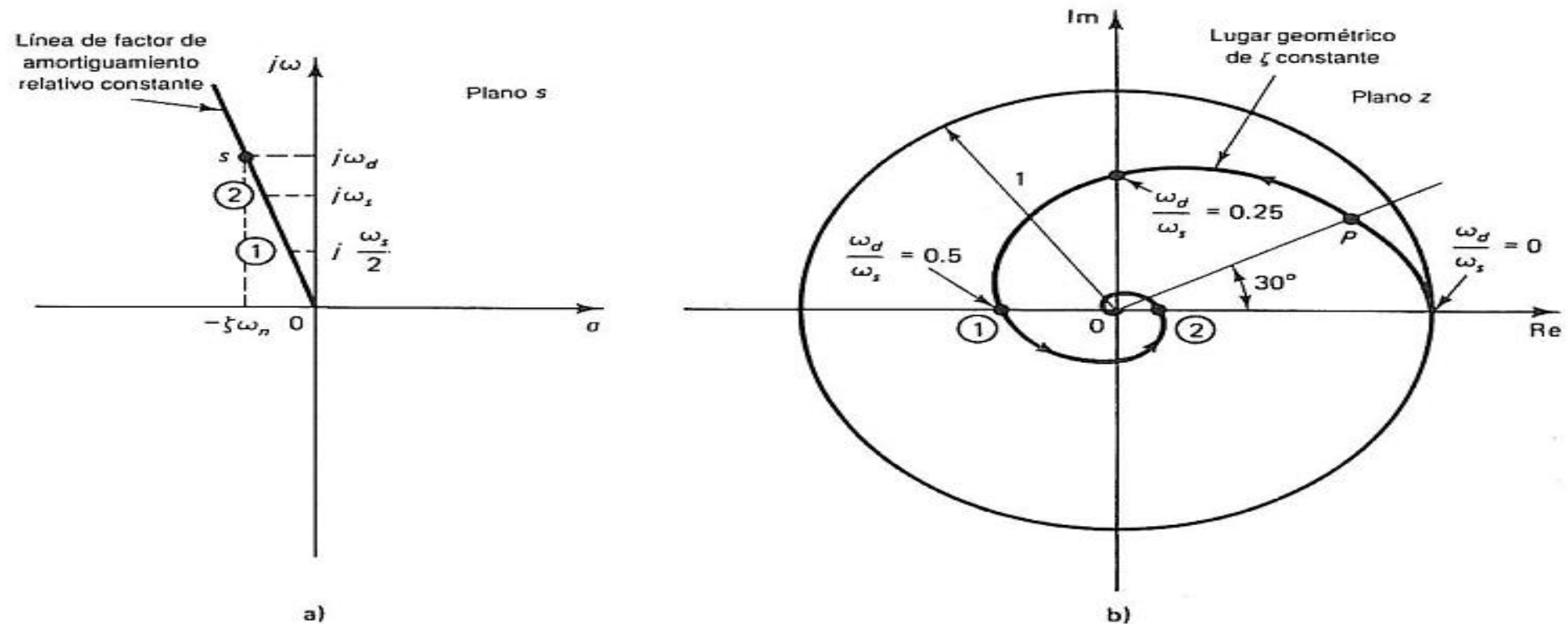


Figura 4-7 a) Linha de fator de amortiguamento relativo constante en el plano s ; b) lugar geométrico correspondiente en el plano z .

Fator de amortecimento constante

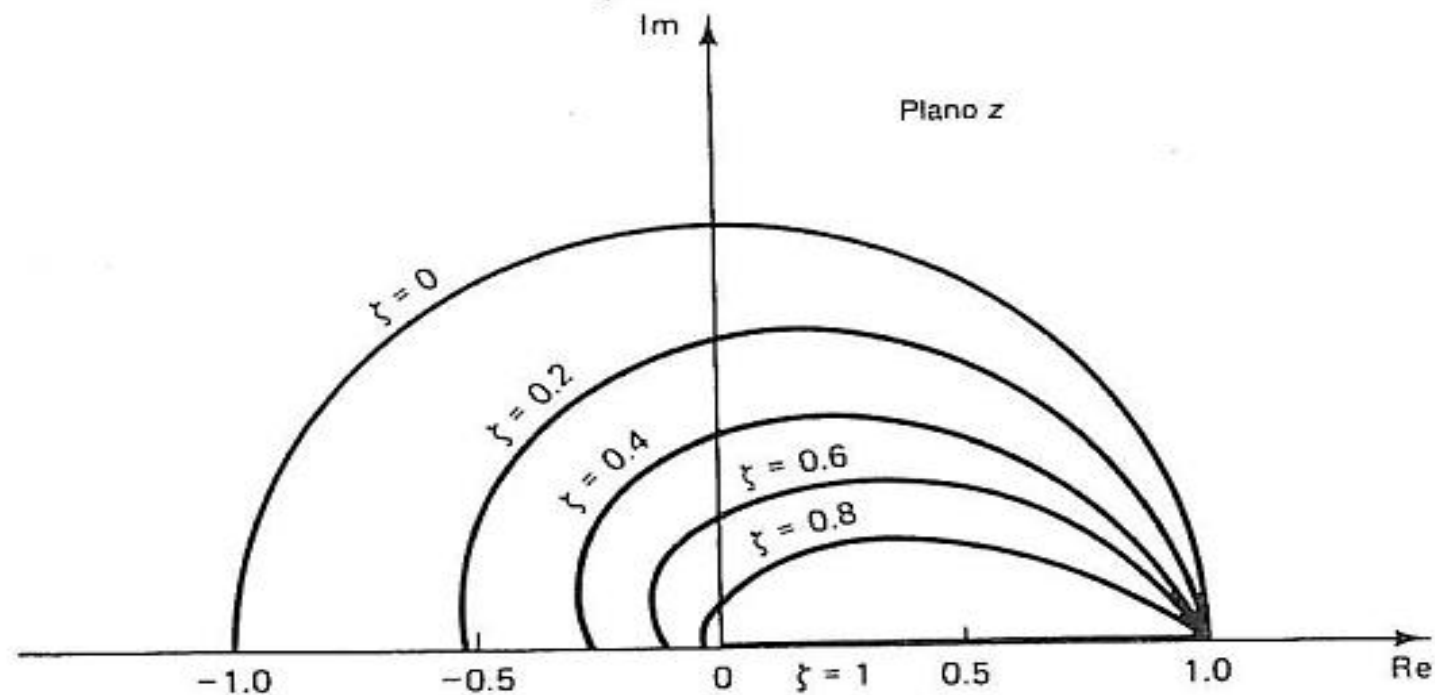
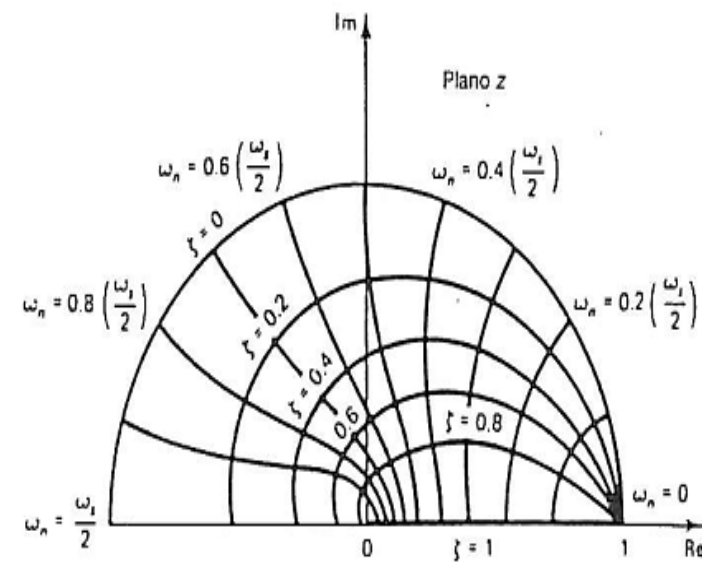
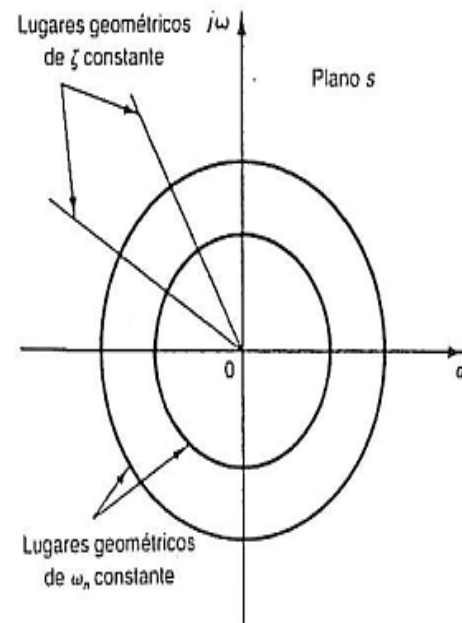
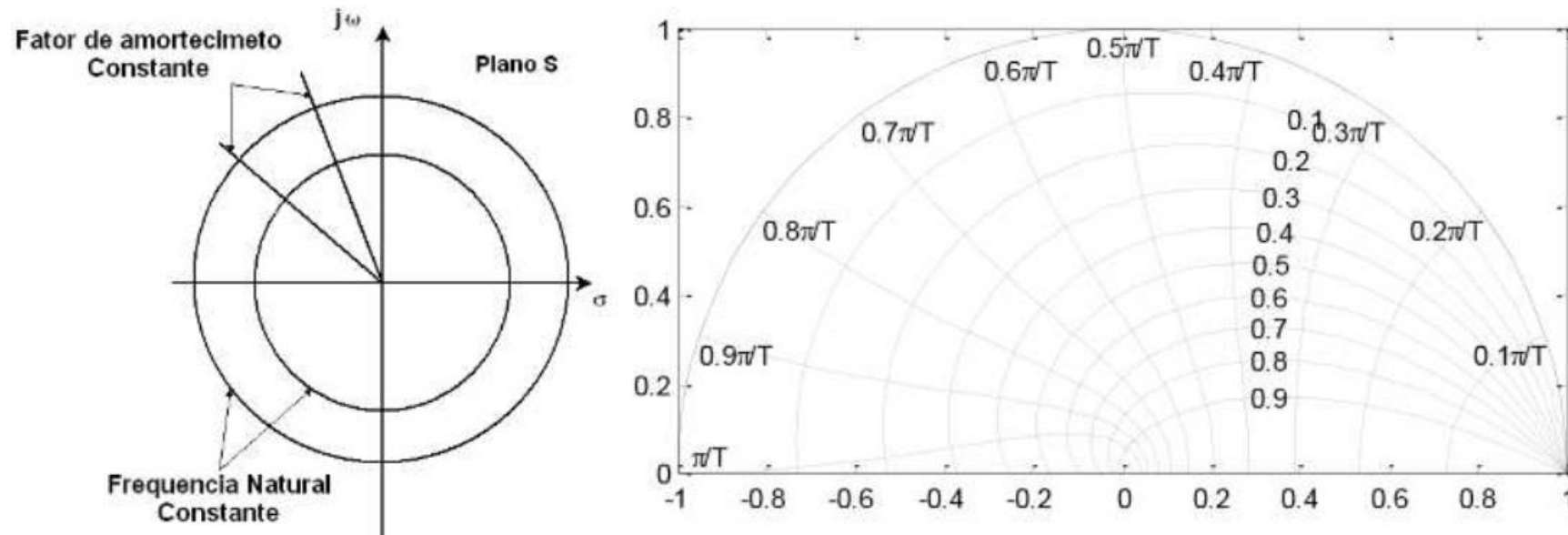


Figura 4-8 Lugares geométricos do fator de amortecimento relativo constante em o plano z .

ζ e ω_n constantes



Plano “s”  Plano “z”



Curvas de frequência natural (ω_n) e fator de amortecimento (ζ) constantes

Exemplo

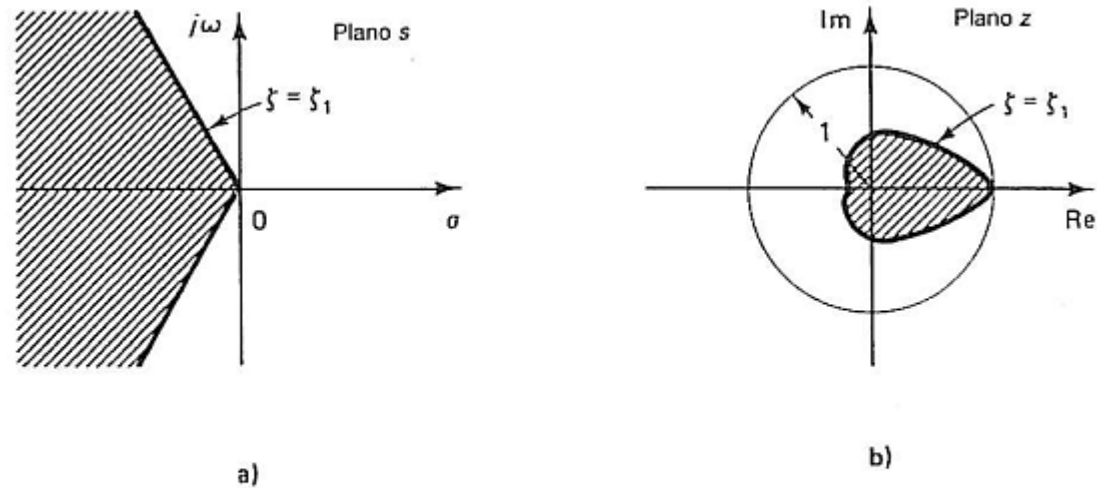


Figura 4-10 a) Región correspondiente a $\zeta > \zeta_1$ en el plano s ; b) región correspondiente a $\zeta > \zeta_1$ en el plano z .

Exemplo

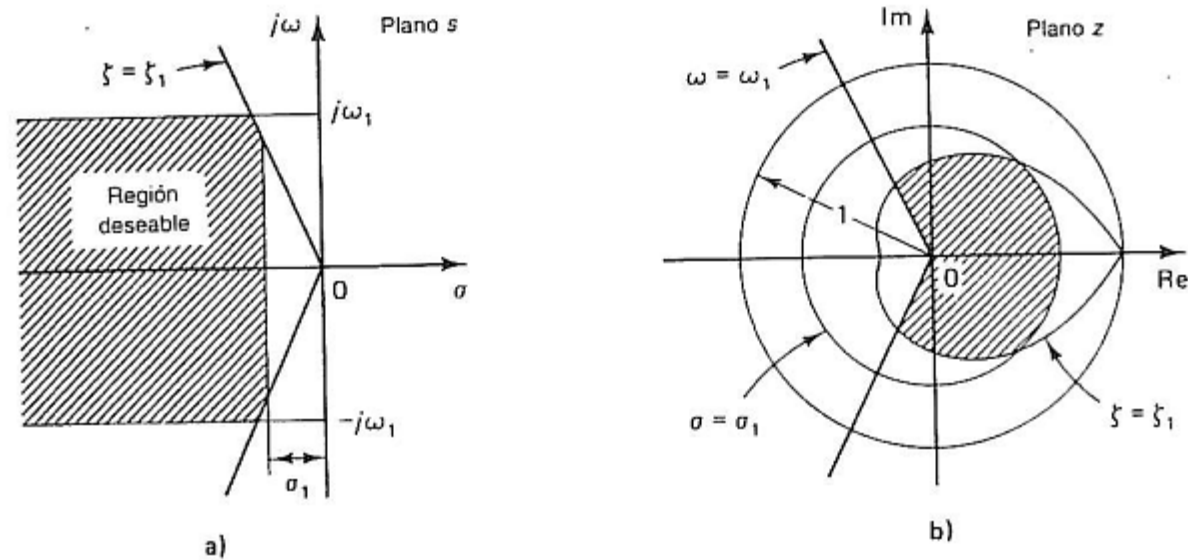


Figura 4-11 a) Una región deseable en el plano s para la localización de los polos en lazo cerrado; b) región correspondiente en el plano z .

Referencias bibliográficas

- O material apresentado foi baseado nos livros
 - K. Ogata, Discrete-time Control Systems, Prentice-Hall, 1987 (Cap. 2 a 4).
 - B. Kuo – Digital Control Systems, 2nd Ed., Saunders HBJ
- OBS: algumas figuras foram retiradas diretamente desses materiais, outras estão disponibilizadas na web e as demais são de autoria própria.